MatchClerk — Project Implementation Documents

Product: MatchClerk (SME 3-Way Invoice Matcher)
Company: Manasflow
Document type: Full implementation pack (PRD · Architecture · Delivery · GTM · Ops)
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1. Executive Brief

1.1 One-liner

MatchClerk is a productized digital employee that extracts line items from supplier invoice PDFs, matches them to open POs and receiving notes, and drafts coded AP entries in QuickBooks Online or Xero for one-click human approval.

1.2 Strategic role

Dimension	Detail
Wedge type	Trojan-horse micro-product
Beachhead ICP	Controllers & Ops Directors at real-estate, specialty contractors, wholesale distributors (15–150 employees)
Competitive gap	Enterprise tools (Coupa, GEP, Zycus) ignore SME PDF → QBO/Xero 3-way match
Time-to-MVP	4–6 weeks
Primary upsell	Full AP digital employee → procurement agent → Enterprise AI Systems

1.3 Non-goals (MVP)

- Full multi-entity ERP (NetSuite, SAP, Dynamics) — Phase 2+
- Autonomous payment execution (ACH/wire) without human approval
- Supplier portal / e-invoicing network (Peppol, etc.)
- Complex multi-currency hedging or tax advisory
- Replacing the general ledger

1.4 Success definition (first 90 days post-launch)

- 5 paying pilot logos
- ≥85% line-item extraction accuracy on customer invoice samples
- ≥70% auto-match rate (PO + receipt) before human touch
- Median time invoice → draft entry ≤ 3 minutes
- NPS ≥ 40 among pilot controllers

2. Product Requirements Document (PRD)

2.1 Problem statement

SME ops and finance teams process hundreds of supplier invoices monthly by manually opening PDFs, re-keying line items into QuickBooks/Xero, and eyeballing POs and receiving docs. This burns 10–20+ hours/week, creates coding errors, and delays close.

2.2 Solution summary

MatchClerk acts as a supervised AP clerk:

1. **Ingest** invoices (email attachment, upload, or shared folder)
2. **Extract** header + line items via vision/LLM pipeline
3. **Match** against open POs and goods receipts (3-way)
4. **Code** GL accounts / classes / jobs using learned rules + history
5. **Draft** bill in QBO/Xero in pending/approval state
6. **Notify** human for one-click approve / edit / reject
7. **Learn** from corrections for that tenant

2.3 Personas

Persona	Role	Primary jobs	Pain
Ava — Controller	Buyer, admin	Close books, control AP accuracy	Manual keying, month-end crunch
Marcus — Ops Manager	Champion	Job costing, vendor reliability	Wrong codes, delayed invoices
Priya — AP Clerk	Daily user	Enter bills, chase docs	Repetitive PDF work
Sam — Owner/CFO (small firm)	Economic buyer	Cost control	Headcount vs. software spend

2.4 Core user flows

Flow A — Email ingest (happy path)

```
Vendor emails invoice PDF to ap@customer.com (or forward address)
  → MatchClerk mailbox watcher captures message + attachment
  → Virus/type scan → store encrypted blob
  → Extraction job runs
  → Matching engine scores PO + receipt candidates
  → Draft bill created in QBO/Xero (or internal pending if low confidence)
  → Slack/email card: Approve | Edit | Reject
  → On Approve: post/open bill in accounting system
  → Audit log + metrics updated
```

Flow B — Manual upload

```
User drops PDF/PNG/ZIP in web app or drag-and-drop
  → Same pipeline from extraction onward
```

Flow C — Exception path

```
Low confidence extraction OR no PO match OR amount variance > threshold
  → Status = Needs Review
  → Side-by-side UI: PDF | extracted fields | PO candidates
  → User corrects → optional "always map vendor X like this"
  → Re-run match → draft
```

2.5 Functional requirements

Must-have (MVP)

ID	Requirement
F1	Upload PDF/PNG invoices (single + batch ZIP)
F2	Email ingest via dedicated forwarding address per tenant
F3	Extract: vendor name, invoice #, invoice date, due date, currency, tax, total, line items (desc, qty, unit price, amount)
F4	Connect QuickBooks Online (OAuth2) — read POs/bills/vendors; write draft bills
F5	Connect Xero (OAuth2) — equivalent read/write
F6	3-way match: Invoice ↔ PO ↔ Receipt (receipt optional if tenant config = 2-way)
F7	Configurable variance tolerances (% and absolute $)
F8	GL/account coding suggestions from vendor history + rules
F9	Human approval queue (web + email magic link; Slack optional MVP+)
F10	Audit log: who approved what, before/after field values
F11	Tenant isolation, encrypted file storage, zero-retention LLM option
F12	Admin: users, tolerances, vendor aliases, GL default rules

Should-have (v1.1, weeks 7–10)

ID	Requirement
F13	Slack interactive approve/reject
F14	Duplicate invoice detection (vendor + invoice # + amount)
F15	Multi-currency basic (FX rate from accounting system)
F16	Job/class/location dimensions for contractors
F17	Vendor statement reconciliation assist
F18	CSV export of match reports

Won’t-have (later)

ID	Requirement
F19	NetSuite / Sage / Exact connectors
F20	Autonomous payment runs
F21	Full procurement (PO creation, RFQ)
F22	Mobile-native app

2.6 Non-functional requirements

ID	Category	Target
NF1	Extraction latency	p95 < 90s per invoice (≤10 pages)
NF2	End-to-end draft latency	p95 < 3 min email → notify
NF3	Availability	99.5% monthly (business hours weighted)
NF4	Security	SOC 2–aligned controls; encryption at rest + in transit
NF5	Tenancy	Hard logical isolation; no cross-tenant model bleed
NF6	Scale (MVP)	50 tenants × 2,000 invoices/tenant/month
NF7	Accessibility	Web UI WCAG 2.1 AA critical flows
NF8	Observability	Structured logs, traces per invoice_id, error budgets

3. User Stories & Acceptance Criteria

Epic E1 — Ingest

US-1.1 As an AP clerk, I upload a PDF so MatchClerk can process it without email setup.
AC: Supported MIME types only; max 25MB; progress indicator; appears in queue < 5s.

US-1.2 As a controller, I forward vendor emails to a unique address so invoices arrive automatically.
AC: Unique `tenant+ap@ingest.manasflow.com` (or similar); SPF/DKIM handled; attachment extracted; non-invoice mail discarded with log.

Epic E2 — Extract

US-2.1 As an AP clerk, I see extracted header + lines side-by-side with the PDF.
AC: All F3 fields editable; confidence badges (high/med/low); page navigation for multi-page PDFs.

US-2.2 As a controller, I need reliable totals.
AC: Line sum + tax ≈ total within $0.05 or flagged; currency symbol normalized.

Epic E3 — Match

US-3.1 As an ops manager, I want invoices matched to the correct PO.
AC: Ranked PO candidates with score; match on vendor + amount + line similarity; manual override.

US-3.2 As a controller, I configure 2-way vs 3-way match.
AC: Tenant setting; if 3-way and no receipt, status = Awaiting Receipt (not draft).

US-3.3 As an AP clerk, I see variance explanations.
AC: UI shows qty/price/tax deltas vs PO; over-tolerance blocks auto-draft.

Epic E4 — Code & draft

US-4.1 As a controller, historical coding is reused.
AC: Last 5 bills for vendor influence default GL; rule engine can pin vendor → account.

US-4.2 As an AP clerk, approve posts a draft bill.
AC: Idempotent write; external_id stored; failure retries with backoff; user sees QBO/Xero deep link.

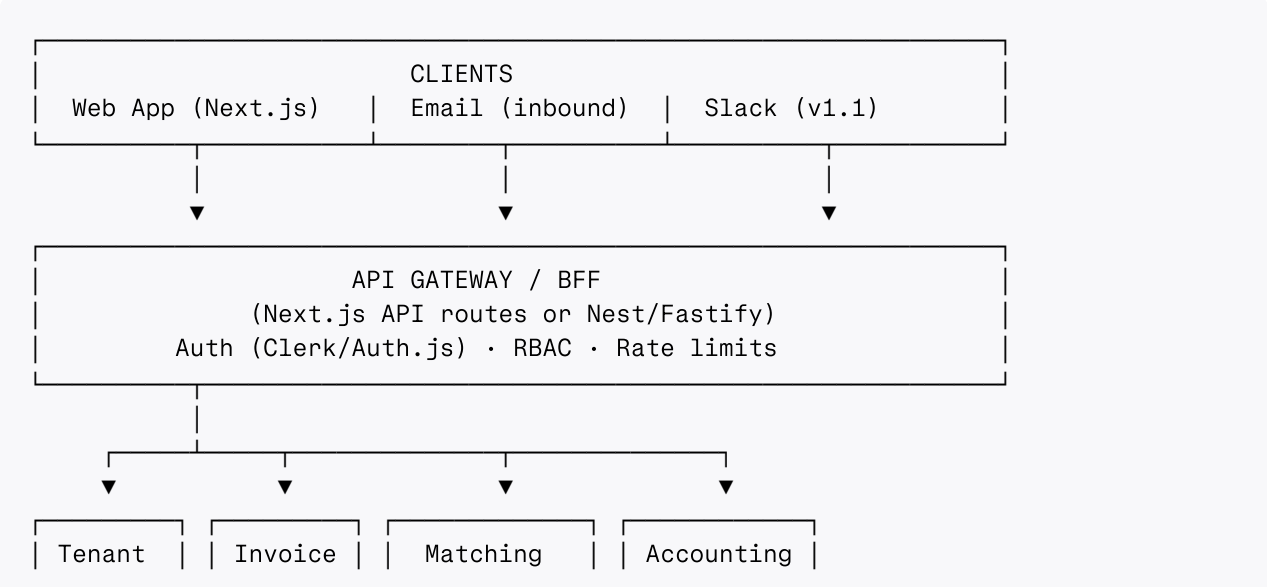
Epic E5 — Govern

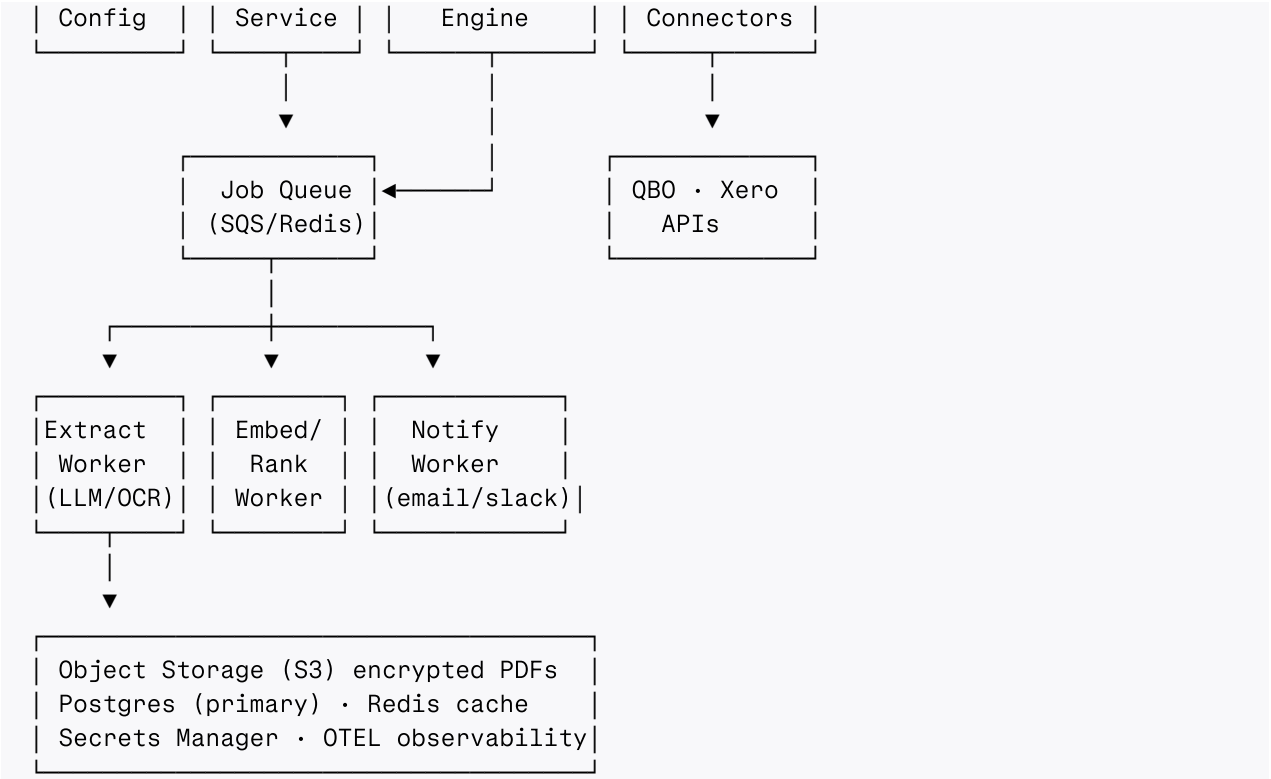
US-5.1 As a controller, I audit every change.
AC: Immutable event log; filter by invoice, user, date; export CSV.

US-5.2 As an owner, data doesn't train public models.
AC: Zero-retention vendor API flags where available; DPA; tenant data deletion within 30 days of churn request.

4. System Architecture

4.1 High-level diagram (logical)





4.2 Recommended stack (Manasflow-aligned)

Layer	Choice	Notes
Frontend	Next.js 15 App Router + TypeScript	Matches agency stack
UI	Tailwind + shadcn/ui	Fast, consistent
Auth	Clerk or Auth.js + Google/Microsoft SSO	SSO matters for finance buyers
API	Next.js route handlers **or** separate Node service	Split if workers need scale
DB	PostgreSQL (Neon/Supabase/RDS)	RLS for tenant isolation
ORM	Prisma or Drizzle	Prefer Drizzle if edge-heavy
Queue	Inngest **or** BullMQ + Redis	Inngest simpler for MVP
Files	S3-compatible (R2/S3) + SSE-KMS	Pre-signed URLs only
LLM	OpenAI GPT-4o / Anthropic Claude (vision)	Abstract behind provider interface
OCR assist	Optional Textract/Document AI fallback	For dense scanned tables
Email inbound	Postmark / SES inbound + webhook	Per-tenant address routing
Email outbound	Resend or Postmark	Approval magic links
Hosting	Vercel (web) + Fly/Render/AWS (workers)	Keep EU region option for MT/EU clients
Monitoring	Sentry + Axiom/Datadog + OpenTelemetry	Trace `invoice_id` end-to-end

4.3 Environment topology

Env	Purpose
`local`	Docker compose: Postgres, Redis, MinIO
`staging`	Sandbox QBO/Xero apps; synthetic invoices
`prod`	Live connectors; stricter rate limits; backups

5. Data Model

5.1 Core entities (logical)

```
Tenant
  id, name, slug, plan, timezone, match_mode (TWO_WAY|THREE_WAY),
  amount_tolerance_pct, amount_tolerance_abs, created_at

User
  id, tenant_id, email, role (OWNER|CONTROLLER|CLERK|AUDITOR),
  auth_subject

AccountingConnection
  id, tenant_id, provider (QBO|XERO), status,
  access_token_enc, refresh_token_enc, realm_id/tenant_id_ext,
  scopes, connected_by, connected_at, last_sync_at

VendorAlias
  id, tenant_id, raw_name, normalized_name, external_vendor_id,
  default_gl_account_id, default_class_id

Invoice
  id, tenant_id, source (EMAIL|UPLOAD|API), status,
  storage_key, sha256, page_count,
  vendor_name_raw, vendor_id, invoice_number, invoice_date, due_date,
  currency, subtotal, tax_total, total, confidence_json,
  external_bill_id, created_at, updated_at

InvoiceLine
  id, invoice_id, line_index, description, quantity, unit_price,
  amount, tax_amount, gl_account_id, class_id, job_id,
  matched_po_line_id, confidence

PurchaseOrder (synced cache)
  id, tenant_id, external_id, vendor_id, po_number, status,
  currency, total, ordered_at, raw_json, synced_at

PurchaseOrderLine
  id, po_id, external_line_id, description, quantity, unit_price,
  amount, qty_received, qty_billed

Receipt (Goods Received)
  id, tenant_id, external_id, po_id, received_at, raw_json
```

```
ReceiptLine
  id, receipt_id, po_line_id, quantity_received

MatchResult
  id, invoice_id, po_id, receipt_id, score, status
  (AUTO|SUGGESTED|MANUAL|REJECTED), variance_json, decided_by, decided_at

Approval
  id, invoice_id, action (APPROVE|EDIT_APPROVE|REJECT),
  actor_user_id, comment, snapshot_json, created_at

AuditEvent
  id, tenant_id, entity_type, entity_id, action, actor_user_id,
  before_json, after_json, ip, created_at

ProcessingJob
  id, tenant_id, invoice_id, type, status, attempts, last_error,
  started_at, finished_at
```

5.2 Invoice status state machine

```
RECEIVED
  → EXTRACTING
  → EXTRACTED | EXTRACTION_FAILED
  → MATCHING
  → MATCHED_AUTO | NEEDS_REVIEW | AWAITING_RECEIPT | DUPLICATE_SUSPECT
  → DRAFT_WRITTEN | DRAFT_FAILED
  → APPROVED | REJECTED
  → POSTED_EXTERNAL (bill open/paid sync optional)
```

Illegal transitions must be rejected server-side.

6. Integrations Specification

6.1 QuickBooks Online

Operation	API	Notes
OAuth2 connect	Intuit OAuth	Separate dev + prod apps
List vendors	`GET /v3/company/{id}/query?query=select * from Vendor`	Cache nightly + on-demand
List POs	PurchaseOrder entity (where available) or Bill + estimates fallback	Confirm app entitlements early
List bills	Bill	Duplicate check
Create draft bill	`POST Bill` with `PrivateNote` + custom field `MatchClerkInvoiceId`	Idempotency key
Accounts list	Account	For GL coding UI

Operation	API	Notes
Classes / Locations	Class, Department	Optional dimensions

Critical spike (Week 1): Confirm PO and Bill line linking behavior for target QBO subscriptions. If PO API is limited on some plans, implement "bill vs open bill + vendor credits" fallback and document ICP constraint.

6.2 Xero

Operation	API	Notes
OAuth2	Xero OAuth 2.0	Tenants (orgs) picker
Contacts	Contacts	Vendor map
POs	PurchaseOrders	Stronger native PO support
Bills	Invoices with `Type=ACCPAY`	Status DRAFT
Accounts	Accounts	GL
Tracking categories	TrackingCategories	Class/job analogue

6.3 Email ingest

1. DNS: `ingest.manasflow.com` MX → provider
2. Address pattern: `{tenant_slug}@ingest.manasflow.com`
3. Webhook → validate signature → parse MIME → enqueue
4. Allowlist optional: only accept from known vendor domains per tenant

6.4 Slack (v1.1)

- Bot in workspace
- Interactive message: Approve / Open review / Reject
- Map Slack user → MatchClerk user

7. AI / Extraction Pipeline

7.1 Pipeline stages

```
1. Normalize PDF → page images (300 DPI) + text layer if present
2. Classify document (invoice vs statement vs other)
3. Header extraction (structured JSON schema)
4. Table/line extraction (schema + repair pass)
5. Arithmetic validation (lines + tax ≈ total)
6. Vendor resolution (alias table + fuzzy + embedding)
```

```
7. Confidence scoring + field-level flags
8. Persist Invoice + InvoiceLines
```

7.2 JSON schema (extractor output)

```
{
  "document_type": "invoice",
  "vendor_name": "string",
  "invoice_number": "string",
  "invoice_date": "YYYY-MM-DD",
  "due_date": "YYYY-MM-DD|null",
  "currency": "ISO-4217",
  "subtotal": 0,
  "tax_total": 0,
  "total": 0,
  "purchase_order_number": "string|null",
  "lines": [
    {
      "description": "string",
      "quantity": 1,
      "unit_price": 0,
      "amount": 0,
      "tax_amount": 0
    }
  ],
  "warnings": ["string"]
}
```

7.3 Prompting & model policy

- Use **strict JSON schema** decoding / tool calling
- Temperature 0–0.2
- Two-pass: extract → validate/repair
- Prefer text layer when high quality; vision for scans
- **Never** send data to model providers that retain for training (org zero-retention agreements)
- PII minimization: strip bank QR noise if not needed for match
- Store only extraction result + hash; raw PDF in encrypted object storage

7.4 Confidence model (simple MVP)

Signal	Weight
Schema valid + arithmetic OK	+0.35
Vendor resolved to known ID	+0.20
Invoice # present & unique	+0.15
PO number found on invoice & exists	+0.15

Signal	Weight
Line count ≥ 1 with amounts	+0.15

- `confidence ≥ 0.85` → auto-advance to match
- `0.60–0.85` → NEEDS_REVIEW but pre-filled
- `< 0.60` → NEEDS_REVIEW hard gate

Tune on pilot samples weekly.

7.5 Evaluation set

Before launch, build golden set:

- 100 invoices across ≥10 vendors
- Mix: digital PDF, scan, multi-page, multi-tax, credit notes
- Metrics: field accuracy, line F1, total exact-match rate

Gate: ≥85% header critical fields exact; ≥80% line amount F1.

8. Matching Engine Rules

8.1 Candidate retrieval

1. If invoice has PO number → exact PO lookup for vendor
2. Else retrieve open POs for resolved vendor (last 180 days)
3. Optional embedding similarity on line descriptions for ranking

8.2 Scoring (MVP heuristic)

```
score =
  0.35 * amount_similarity(invoice.total, po.remaining) +
  0.25 * date_proximity(invoice_date, po_date) +
  0.25 * line_overlap(invoice_lines, po_lines) +
  0.15 * po_number_bonus
```

8.3 Three-way logic

Condition	Result
PO matched AND receipt qty covers invoice qty (within tol)	MATCHED_AUTO eligible
PO matched AND receipt missing AND mode=THREE_WAY	AWAITING_RECEIPT
PO matched AND receipt missing AND mode=TWO_WAY	MATCHED_AUTO eligible
Amount variance > tolerance	NEEDS_REVIEW
Multiple POs high score	NEEDS_REVIEW (disambiguate)

Condition	Result
No PO	NEEDS_REVIEW (allow “non-PO bill” if permission enabled)

8.4 Tolerances (tenant defaults)

Setting	Default
`amount_tolerance_pct`	1.0%
`amount_tolerance_abs`	$2.00 (or 2 EUR)
`qty_tolerance`	0 (exact) unless tenant allows

Use **max(pct, abs)** rule: variance allowed if within either threshold (configurable to require both).

8.5 Coding rules priority

1. Explicit VendorAlias default GL
2. Exact prior bill line description → account map
3. Fuzzy prior description
4. Tenant default expense account
5. Leave blank → force review

9. Security, Privacy & Compliance

9.1 Control checklist

Control	Implementation
AuthN	SSO + MFA enforced for OWNER/CONTROLLER
AuthZ	RBAC on every API; tenant_id from session only
Encryption in transit	TLS 1.2+
Encryption at rest	DB encryption + S3 SSE-KMS
Secrets	Vault/Secrets Manager; never in env git
Token storage	Accounting OAuth tokens encrypted (AES-GCM KMS key)
Network	Private workers; least-privilege IAM
Logging	No full PDF content in logs; redacted tokens
Backups	Daily encrypted; 30-day PITR
Retention	Configurable; default PDF 24 months; delete on request
DPA	Standard Manasflow DPA + subprocessors list

Control	Implementation
Regions	Offer EU storage for EU tenants if required

9.2 Threat model (top risks)

1. Stolen QBO/Xero OAuth tokens → bill fraud
2. Cross-tenant file access via IDOR
3. Prompt injection via invoice PDF text
4. Email spoofing to ingest address

Mitigations: token encryption + rotation; UUID + RLS + integration tests for IDOR; treat extracted text as untrusted; optional vendor-domain allowlists + DMARC on outbound.

9.3 Prompt injection posture

- Extractor outputs structured fields only; never execute tool calls from invoice content
- Matching uses DB state, not free-form model actions
- Approval always human for external write in MVP

10. Delivery Plan & Timeline

10.1 Overview (6-week MVP)

Week	Theme	Exit criteria
W1	Foundations + spikes	Repo, auth, tenancy, QBO/Xero OAuth spike green
W2	Ingest + storage + extract v0	Upload → JSON extract on 20 sample PDFs
W3	Matching + review UI	Side-by-side review; tolerances; statuses
W4	Accounting write path	Draft bill in QBO **and** Xero sandbox
W5	Email ingest + approvals + audit	Forward-to-process; magic-link approve; audit log
W6	Hardening + pilot readiness	Golden-set metrics met; 1 internal dogfood + 2 design partners

10.2 Phase roadmap post-MVP

Phase	Window	Scope
v1.1	W7–10	Slack, duplicates, dimensions (job/class), CSV reports
v1.2	W11–14	Multi-entity, credit notes, statement assist
v2.0	W15–20	NetSuite connector, payment-run prep (still human-approved)
Upsell pack	Parallel	AP exception agent, vendor onboarding agent

10.3 Team shape (lean)

Role	Allocation
Tech lead / full-stack	1.0
Full-stack / integrations	1.0
AI/extraction engineer	0.5–1.0
Design (UI/UX)	0.25–0.5
Founder / GTM / pilot success	0.5

11. Engineering Work Breakdown

11.1 Epic → tickets (MVP)

A. Platform

- [ ] Monorepo or app scaffold (Next.js + workers)
- [ ] CI (lint, typecheck, test, migrate)
- [ ] Postgres schema + migrations
- [ ] Tenant middleware + RLS policies
- [ ] RBAC roles
- [ ] File upload pre-signed URL flow
- [ ] Encryption helpers for tokens

B. Auth & admin

- [ ] Sign-up / invite users
- [ ] Tenant settings page (tolerances, match mode)
- [ ] Vendor alias admin CRUD
- [ ] GL default rules UI

C. Extraction

- [ ] PDF normalize service
- [ ] LLM provider interface + OpenAI/Anthropic adapters
- [ ] Schema validation + arithmetic checks
- [ ] Confidence scorer
- [ ] Golden-set eval script (CI nightly optional)

D. Matching

- [ ] PO/receipt sync jobs per connector
- [ ] Candidate retrieval + scorer
- [ ] Variance calculator
- [ ] Status transitions

E. Connectors

- [ ] QBO OAuth + token refresh
- [ ] QBO vendor/PO/bill/account sync
- [ ] QBO draft bill create (idempotent)
- [ ] Xero OAuth + token refresh
- [ ] Xero contact/PO/ACCPAY draft create
- [ ] Connection health dashboard

F. Product UI

- [ ] Inbox / queue list (filters by status)
- [ ] Invoice detail: PDF viewer + field editors
- [ ] Match panel: PO candidates + variance
- [ ] Approve / reject actions
- [ ] Empty states + onboarding checklist

G. Notifications

- [ ] Email: “Invoice ready for approval”
- [ ] Magic link token (short TTL, single use)
- [ ] Failure alerts to tenant admins

H. Observability & ops

- [ ] Sentry + structured logs
- [ ] Per-invoice trace ID
- [ ] Admin internal dashboard (job failures)
- [ ] Runbooks (see Appendix)

11.2 Definition of Done (every ticket)

- Types pass; unit/integration tests where logic is non-trivial
- Tenant isolation covered for data access paths
- Metrics/logs added for failures
- UX copy reviewed for non-finance jargon where possible
- No secrets in git

12. QA, Testing & Launch Criteria

12.1 Test pyramid

Layer	Focus
Unit	Scorer, tolerances, confidence, state machine
Integration	Connector sandboxes, ingest webhook, DB RLS
E2E	Upload → review → approve → bill exists in sandbox
Eval	Extraction golden set
Security	IDOR suite, authz matrix, token encryption smoke

12.2 Launch checklist (go / no-go)

Gate	Threshold
Extraction golden set	≥85% critical header accuracy
E2E sandbox QBO	10/10 bills drafted correctly
E2E sandbox Xero	10/10 bills drafted correctly
IDOR tests	0 critical findings
p95 process time	< 3 minutes on staging samples
Backup restore drill	Successful within last 14 days
DPA + privacy page	Published
Support mailbox + runbooks	Live
Pricing page / contract template	Ready

12.3 Pilot success instrumentation

Track per tenant:

- invoices_received
- extraction_success_rate

- auto_match_rate
- median_human_handle_seconds
- approval_rate
- hours_saved_estimate (baseline survey × volume)
- weekly active clerks

13. Pricing, Packaging & Success Metrics

13.1 Packaging (recommended)

Plan	Price	Includes	Limits
Starter	$800/mo	1 entity, QBO **or** Xero, email ingest, web approve	500 invoices/mo
Growth	$1,500/mo	Both connectors, Slack (v1.1), jobs/classes, 3 users	2,000 invoices/mo
Scale	Custom	Multi-entity, SSO, MSA, MSA security review	5,000+

Overage: $0.60–$1.00 per invoice over cap (confirm margin vs LLM cost).

Implementation fee (optional): $1,500 one-time for historical vendor alias + GL rule setup (waive for design partners).

13.2 Unit economics targets

Metric	Target
COGS/invoice (LLM+OCR+storage)	< $0.15 blended
Gross margin	≥ 75%
Pilot payback	< 2 months CAC
Logo churn (annual)	< 15%

13.3 North-star metric

Approved invoices per week with < 60 seconds human touch time.

Secondary: auto-match rate, expansion revenue to full AP agent within 6 months.

14. Pilot Playbook

14.1 Ideal design partner profile

- 200–2,000 supplier invoices/month
- QuickBooks Online or Xero (not desktop-only)
- 1 engaged controller + 1 AP clerk
- Willing to share 30–50 historical PDFs under NDA
- Real estate ops, specialty contractor, or distributor

14.2 30-day pilot structure

Day	Activity
0	Contract + DPA + security questionnaire
1	Kickoff; connect QBO/Xero; set tolerances
2–3	Upload historical sample; tune vendor aliases
4	Enable email ingest; parallel-run (MatchClerk drafts, human still posts manually if needed)
5–20	Weekly office hours; accuracy review Friday
21	Baseline vs actual hours survey
28–30	Go/no-go to paid; case study interview

14.3 Pilot commercial terms

- $0–500/mo pilot fee (or free for case study rights)
- Convert at Starter/Growth with 2-month lock
- Require measurable baseline: hours/week on AP entry

14.4 Onboarding checklist (customer-facing)

1. Connect accounting system
2. Choose 2-way or 3-way match
3. Set tolerance
4. Invite clerk
5. Forward test invoice
6. Approve first draft
7. Add top 20 vendor aliases

15. Upsell Path to Full Digital Employees

MatchClerk is the wedge. Map expansion deliberately.

```
MatchClerk (invoice match + draft)
    ↓ lands in AP inbox daily
AP Exception Agent
    · vendor statement reconcile
    · duplicate & fraud flags
    · payment-run prep packs
    ↓
Procurement Digital Employee
    · PO drafting from approved requests
    · vendor scorecards (responsiveness, variance)
    · catalog price watch (MarginGuard attach)
    ↓
Enterprise AI Systems
    · job costing sync (contractors)
    · cross-dept hub (AP ⇔ inventory ⇔ project ops)
    · custom multi-agent retainers
```

Trigger events for upsell conversation

Signal	Offer
>1,000 invoices/mo	Growth → Scale + exception agent
High AWAITING_RECEIPT volume	Receiving workflow agent
Contractor job codes heavily used	Job-costing / project ops agents
Asks for PO creation	Procurement employee
Multiple entities	Enterprise Systems engagement

16. Risks & Mitigations

Risk	Impact	Likelihood	Mitigation
QBO PO API limitations on lower SKUs	High	Med	Week-1 spike; position Xero-first if needed; 2-way bill match fallback
Extraction fails on ugly scans	High	Med	Textract fallback; human review UX excellence
LLM cost spikes	Med	Med	Caching; smaller model for clean digital PDFs; batch
Accounting buyers demand SOC2	High	High	Start SOC2 Type 1 parallel; share security whitepaper
Slow sales cycle (“wait until FY”)	Med	Med	Pilot pricing; month-to-month; hours-saved ROI one-pager
Scope creep into full AP suite	High	High	Ruthless MVP cut list; park ideas in v1.1+ backlog
Connector breaking API changes	Med	Low	Contract tests; status page; multi-provider abstraction

17. Appendix

17.1 Sample API routes (illustrative)

```
POST   /api/invoices/upload
POST   /api/webhooks/email-inbound
GET    /api/invoices?status=
GET    /api/invoices/:id
PATCH  /api/invoices/:id          # field corrections
POST   /api/invoices/:id/match    # re-run match
POST   /api/invoices/:id/approve
POST   /api/invoices/:id/reject
GET    /api/connections
POST   /api/connections/qbo/start
GET    /api/connections/qbo/callback
POST   /api/connections/xero/start
GET    /api/settings
PATCH  /api/settings
GET    /api/audit?invoice_id=
```

17.2 Sample approval email copy

> **Subject:** Invoice {{vendor}} · {{invoice_number}} · {{total}} ready
> MatchClerk matched this invoice to PO {{po_number}} (score {{score}}).
> Variance: {{variance_summary}}
> [Approve] [Review] [Reject]
> — MatchClerk by Manasflow

17.3 Runbooks (titles)

1. OAuth token refresh failures
2. Extraction provider outage failover
3. Stuck jobs in EXTRACTING
4. Duplicate bill write prevention
5. Tenant offboarding & data deletion
6. Suspected fraudulent invoice escalation

17.4 Week-1 technical spike checklist

- [ ] Create Intuit developer app (sandbox)
- [ ] Create Xero developer app (demo company)
- [ ] Prove: read vendors, read POs, create DRAFT bill, idempotent re-write
- [ ] Prove: PDF vision extract on 10 real sample invoices (anonymized)
- [ ] Decide queue product (Inngest vs BullMQ)
- [ ] Decide EU vs US primary region

- [ ] Draft threat model one-pager

17.5 Open decisions log

ID	Decision	Options	Owner	Due
D1	Auth vendor	Clerk vs Auth.js	Eng	W1
D2	Queue	Inngest vs BullMQ	Eng	W1
D3	Primary LLM	OpenAI vs Anthropic default	AI	W1
D4	PO-less bills allowed?	Yes w/ permission vs never MVP	Product	W2
D5	Pilot price	Free vs $500	Founder	W2

17.6 Glossary

Term	Meaning
2-way match	Invoice ↔ PO
3-way match	Invoice ↔ PO ↔ Receipt
Digital employee	Supervised AI agent productized for one job
GL coding	Assigning expense account (and dimensions)
Golden set	Labeled evaluation invoices
Idempotency key	Prevents duplicate bill creation on retry

Document control

Version	Date	Author	Changes
1.0	2026-08-07	Manasflow Product Strategy	Initial MatchClerk implementation pack

Next actions for the team

1. Resolve open decisions D1–D5 in a 45-minute kickoff.
2. Run Week-1 spikes (QBO/Xero + extraction).
3. Recruit 2 design partners from real-estate / contractor network.
4. Stand up repo + staging environment day 1.
5. Schedule Friday accuracy reviews from Week 2 onward.

End of MatchClerk Project Implementation Documents.